

Unit 7: Network Security (5 Hours)

Network security is crucial for protecting data, ensuring privacy, and maintaining trust in communication systems. This unit covers the fundamental concepts, algorithms, and tools used to secure networks.

7.1 A Model for Network Security

A **Network Security Model** outlines how security is implemented in a communication network to ensure **confidentiality**, **integrity**, and **availability** of data.

Key Elements:

1. **Sender:** Initiates communication and sends data.
2. **Receiver:** Receives the transmitted data.
3. **Transmission Medium:** The channel over which the data is transmitted.
4. **Security Mechanisms:** Tools and algorithms (e.g., encryption, authentication).
5. **Security Policies:** Rules defining security requirements.

Core Objectives of Network Security:

- **Confidentiality:** Prevent unauthorized access.
 - **Integrity:** Ensure data remains unaltered during transmission.
 - **Availability:** Ensure network services are accessible when needed.
 - **Authentication:** Verify the identity of communicating entities.
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7.2 Principles of Cryptography: Symmetric Key and Public Key

Cryptography is the science of securing communication by transforming data into an unreadable format.

1. Symmetric Key Cryptography:

- **Key Concept:** Same key is used for encryption and decryption.
- **Algorithm Example:** AES, DES
- **Advantages:** Fast and efficient for large data.
- **Disadvantages:** Key distribution is challenging.

Example:

- Plaintext → Encryption (Key) → Ciphertext → Decryption (Same Key) → Plaintext

2. Public Key Cryptography (Asymmetric):

- **Key Concept:** Uses a pair of keys: Public Key (for encryption) and Private Key (for decryption).
- **Algorithm Example:** RSA, ECC (Elliptic Curve Cryptography)
- **Advantages:** Secure key distribution.
- **Disadvantages:** Slower compared to symmetric algorithms.

Example:

- Public Key → Encryption → Ciphertext → Private Key → Decryption → Plaintext

7.3 Public Key Algorithm — RSA

RSA (Rivest–Shamir–Adleman) is one of the most widely used public-key cryptography algorithms.

Working Principle:

1. **Key Generation:** Generate public and private keys using large prime numbers.
2. **Encryption:** Use the recipient's public key to encrypt data.
3. **Decryption:** Use the recipient's private key to decrypt data.

Advantages:

- Strong security due to large key size.
- Supports secure key exchange.

Applications:

- Digital signatures
- SSL/TLS protocols

Formula:

- Encryption: $C = M^e \bmod N$
- Decryption: $M = C^d \bmod N$

Where:

- $N = p \times q$ (Product of two primes)
- e : Public exponent
- d : Private exponent

7.4 Digital Signature Algorithm (DSA)

DSA is used to ensure **authenticity**, **integrity**, and **non-repudiation** of digital messages.

Key Components:

1. **Key Generation:** Generate public and private keys.
2. **Signing:** Use the private key to sign a hash of the message.
3. **Verification:** Verify the signature using the public key.

Advantages:

- Ensures message authenticity.
- Provides non-repudiation.

Applications:

- Secure emails
- Software distribution

Process:

1. Hash the message.
 2. Sign the hash with a private key.
 3. Send the message and signature.
 4. Receiver verifies the signature with the sender's public key.
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7.5 Communication Security: IPSec, VPN, Firewalls, Wireless Security

1. IPSec (Internet Protocol Security)

- **Purpose:** Secure communication over IP networks.
- **Protocols:** AH (Authentication Header), ESP (Encapsulating Security Payload)
- **Modes:** Transport Mode, Tunnel Mode

Applications: VPNs, secure remote access.

2. VPN (Virtual Private Network)

- **Purpose:** Create a secure encrypted connection over an insecure network (e.g., the internet).
- **Types:** Site-to-Site VPN, Remote Access VPN
- **Protocols:** OpenVPN, L2TP/IPSec

Applications: Remote work, secure data sharing.

3. Firewalls

- **Purpose:** Control incoming and outgoing traffic based on security rules.
- **Types:** Hardware Firewall, Software Firewall
- **Techniques:** Packet Filtering, Stateful Inspection

Applications: Prevent unauthorized access, block malware.

4. Wireless Security

- **Purpose:** Secure wireless networks (Wi-Fi).
- **Protocols:** WPA3, WPA2, WEP
- **Techniques:** MAC Address Filtering, Disable SSID Broadcast

Best Practices:

- Use strong passwords.
 - Enable WPA3 encryption.
 - Disable default admin credentials.
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Conclusion:

Understanding network security principles, cryptographic algorithms, and tools like IPSec, VPNs, and Firewalls is essential for building secure communication systems. These technologies collectively protect data from unauthorized access, tampering, and attacks.

